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LANGCHAIN

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1. Introduction to LangChain

LangChain is an open-source framework designed to help developers build applications powered by Large Language Models (LLMs) like OpenAI GPT or HuggingFace models.

In simple terms, LangChain helps us **connect LLMs with tools, memory, data sources, and workflows** to build intelligent applications.

Why LangChain is Important

Modern LLMs are powerful but have limitations:

- They cannot access real-time data directly
- They don't remember previous conversations well

- They cannot interact with external tools by default

LangChain solves these problems by providing a **modular framework** to build LLM-based systems.

Problems It Solves

- Prompt orchestration
- Multi-step reasoning (chains)
- Memory handling
- Tool and API integration
- Agent-based decision making

2. Core Components of LangChain

1. LLMs and Chat Models

LLMs are the core engines (like GPT-4, Claude, etc.).

Code Example

```
from langchain_openai import ChatOpenAI
```

```
llm = ChatOpenAI(model="gpt-4", temperature=0.7)

response = llm.invoke("Explain LangChain in simple terms")
print(response.content)
```

2. Prompt Templates

Used to structure input prompts dynamically.

Code Example

```
from langchain.prompts import PromptTemplate
```

```
template = PromptTemplate(
    input_variables=["topic"],
    template="Explain {topic} in simple terms for beginners"
)

print(template.format(topic="LangChain"))
```

3. Chains

Chains connect multiple steps together.

Code Example

```
from langchain.chains import LLMChain
from langchain_openai import ChatOpenAI
from langchain.prompts import PromptTemplate
```

```
llm = ChatOpenAI(model="gpt-4")

prompt = PromptTemplate.from_template("Explain {topic}")

chain = LLMChain(llm=llm, prompt=prompt)

result = chain.run("LangChain")
print(result)
```

4. Memory

Memory helps LLMs remember past conversations.

Code Example

```
from langchain.memory import ConversationBufferMemory
from langchain.chains import ConversationChain
from langchain_openai import ChatOpenAI
```

```
llm = ChatOpenAI(model="gpt-4")

memory = ConversationBufferMemory()

conversation = ConversationChain(
    llm=llm,
    memory=memory
)

print(conversation.predict(input="Hi, I am John"))
print(conversation.predict(input="What is my name?"))
```

5. Agents

Agents decide which tool to use dynamically.

Code Example

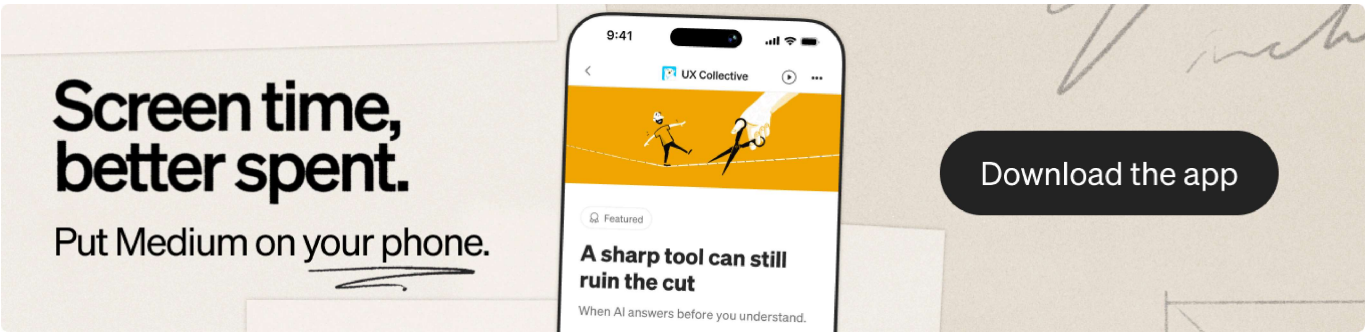
```
from langchain.agents import initialize_agent, Tool
from langchain_openai import ChatOpenAI
```

```
def calculator(x):
    return eval(x)
```

```
tools = [
    Tool(name="Calculator", func=calculator, description="Performs
math")
]

llm = ChatOpenAI(model="gpt-4")

agent = initialize_agent(tools, llm, agent="zero-shot-react-
description")

print(agent.run("What is 25 * 4?"))
```

6. Tools

Tools are external functions like search, calculators, APIs.



Example: Calculator, Search engine, database query system.

7. Document Loaders & Vector Stores

Used for RAG (Retrieval Augmented Generation).

Code Example

```
from langchain.document_loaders import TextLoader
```

```
loader = TextLoader("data.txt")
docs = loader.load()
```

```
print(docs[0].page_content)
```

Vector stores convert text into embeddings for search.

3. Architecture Explanation

Flow of LangChain System:

```
User Input
   ↓
Prompt Template
   ↓
LLM (GPT / HuggingFace)
   ↓
Chain (processing logic)
   ↓
Agent (optional decision making)
   ↓
Tools / APIs
   ↓
Final Output
```

Architecture Diagram (for submission)

You can draw this in diagram tool:

```
User
 ↓
Prompt Template
 ↓
LLM (GPT)
 ↓
Chain
 ↓
```

```
Memory / Agent
 ↓
Tools / APIs / Vector DB
 ↓
Response
```

4. Hands-on Code Examples

Basic LLM Call

```
from langchain_openai import ChatOpenAI
```

```
llm = ChatOpenAI(model="gpt-4")
print(llm.invoke("What is AI?").content)
```

PromptTemplate + Chain

```
from langchain.prompts import PromptTemplate
from langchain.chains import LLMChain
from langchain_openai import ChatOpenAI
```

```
llm = ChatOpenAI(model="gpt-4")

prompt = PromptTemplate.from_template("Explain {topic}")

chain = LLMChain(llm=llm, prompt=prompt)

print(chain.run("Machine Learning"))
```

Agent with Tool

```
from langchain.agents import initialize_agent, Tool
from langchain_openai import ChatOpenAI
```

```
def multiply(x):
    return str(eval(x))

tools = [
    Tool(name="Multiplier", func=multiply, description="Math
operations")
]

llm = ChatOpenAI(model="gpt-4")

agent = initialize_agent(tools, llm, agent="zero-shot-react-
description")

print(agent.run("What is 12 * 8?"))
```

Memory Example

```
from langchain.memory import ConversationBufferMemory
from langchain.chains import ConversationChain
from langchain_openai import ChatOpenAI
```

```
llm = ChatOpenAI(model="gpt-4")

memory = ConversationBufferMemory()

chat = ConversationChain(llm=llm, memory=memory)

print(chat.predict(input="My name is Alex"))
print(chat.predict(input="What is my name?"))
```

5. Real-World Use Cases

1. AI Chatbot for Customer Support

- Problem: Manual customer replies are slow
- Solution: LangChain chatbot with memory
- Components: LLM + Memory + PromptTemplate

2. Document Q&A System

- Problem: Searching large PDFs manually
- Solution: RAG system using vector DB
- Components: Document Loader + Embeddings + Retriever

3. AI Research Assistant

- Problem: Collecting data from multiple sources
- Solution: Agent-based system using tools
- Components: Agent + Tools + LLM

6. Advantages and Limitations

Advantages

- Modular architecture
- Easy integration with APIs
- Fast prototyping of AI apps
- Supports agents and tools

Limitations

- Higher latency due to multiple steps
- Debugging is complex
- API cost can be high
- Requires good prompt engineering

When NOT to use LangChain

- Simple chatbot apps
- Static rule-based systems
- Low-latency production systems

7. Conclusion

LangChain is a powerful framework for building LLM-powered applications. It helps developers connect models with tools, memory, and external data sources.

Key takeaways:

- Understand LLM orchestration
- Build modular AI pipelines
- Use agents and memory effectively

Future Scope:

- LangGraph for advanced workflows
- Multi-agent AI systems

- Autonomous AI applications

Langchain Python Programming Generative Ai Tools

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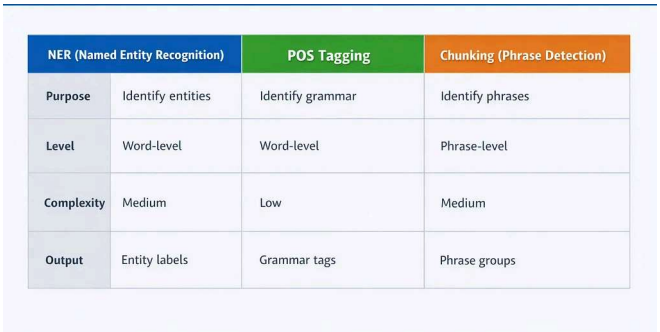

	NER (Named Entity Recognition)	POS Tagging	Chunking (Phrase Detection)
Purpose	Identify entities	Identify grammar	Identify phrases
Level	Word-level	Word-level	Phrase-level
Complexity	Medium	Low	Medium
Output	Entity labels	Grammar tags	Phrase groups

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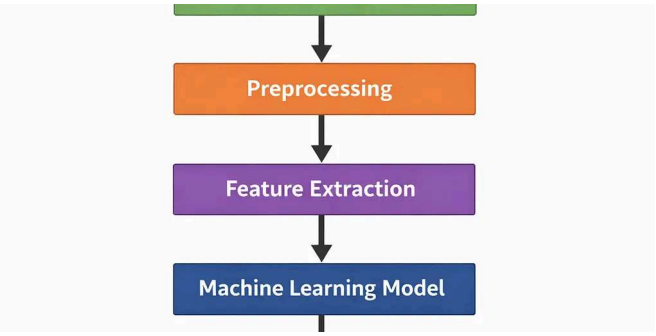

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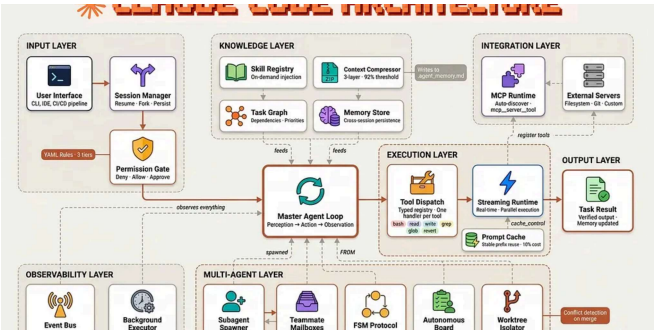

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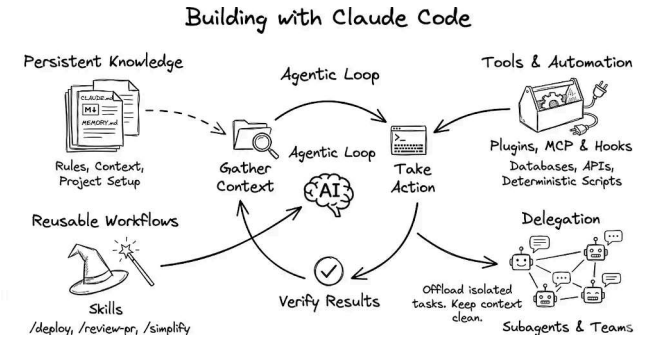

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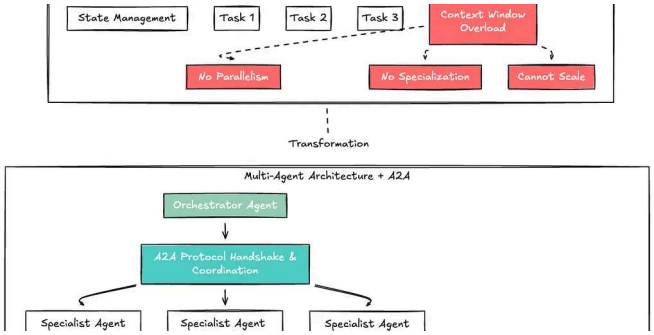

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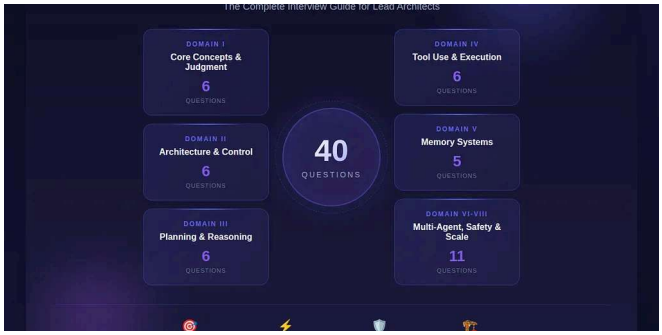

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